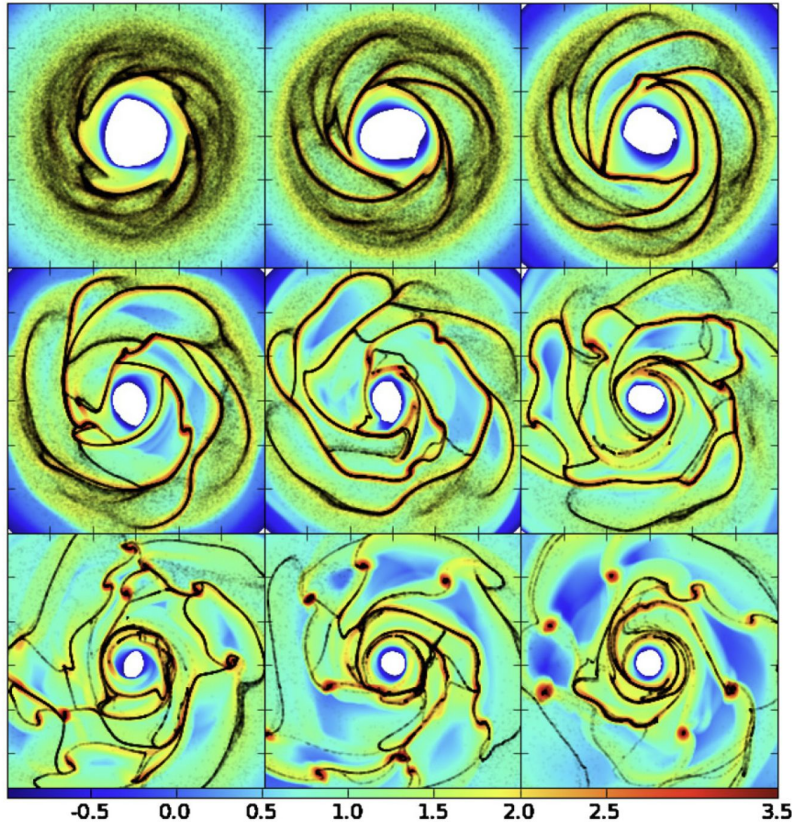


Dust Dynamics in Gravitational Instabilities

PFITS; SETI Institute
June 8, 2026
Hans Baehr (MPIA, Heidelberg)

Dust and Gravitational Instabilities



Boley et al. 2010

Simulations with dust and self-gravity were rare

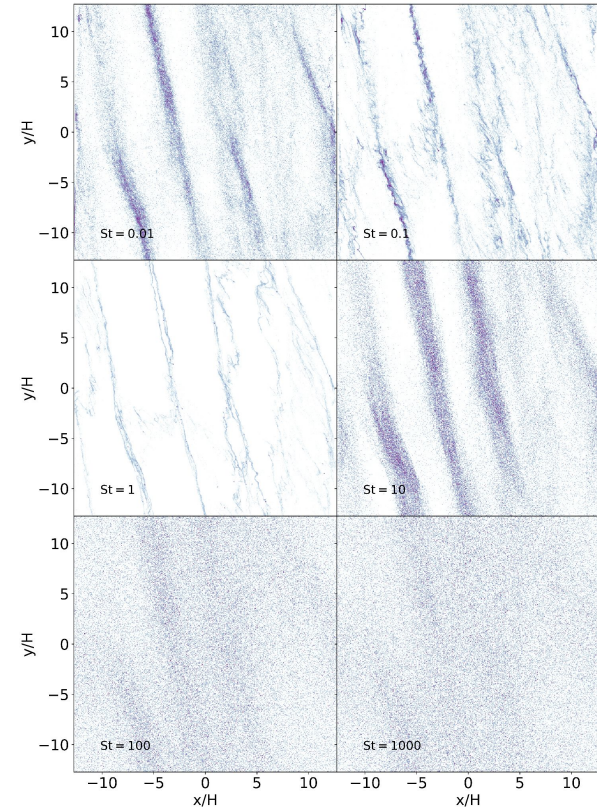
Focused on gas fragmentation

What about gravitoturbulence?

Efficient Radial Drift

Dust of all sizes maintain some non-axisymmetric structure

$$\theta \sim 10^\circ - 14^\circ$$



Baehr & Zhu (2021a)

Efficient Radial Drift

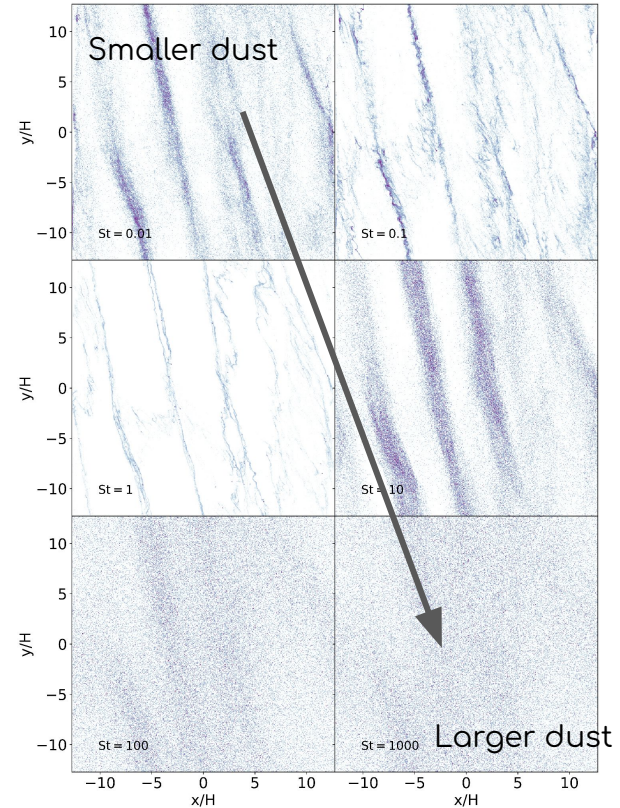
Dust of all sizes maintain some non-axisymmetric structure

$$\theta \sim 10^\circ - 14^\circ$$

Small dust: dominated by drag interactions

Larger dust:
gravitational attraction
and stirring due to gas

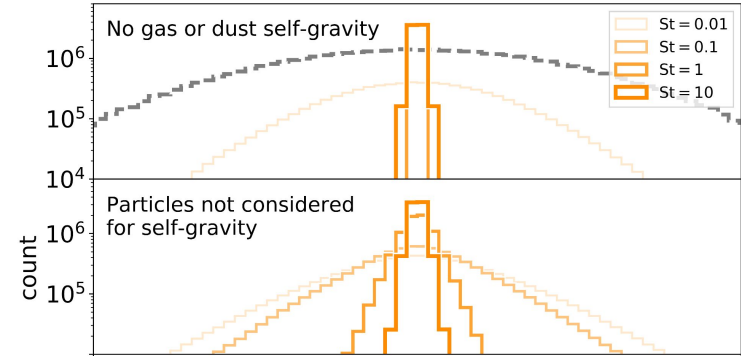
$$\frac{Q}{St} < 1$$



Baehr & Zhu (2021a)

Efficient Settling

Settling versus turbulent diffusion



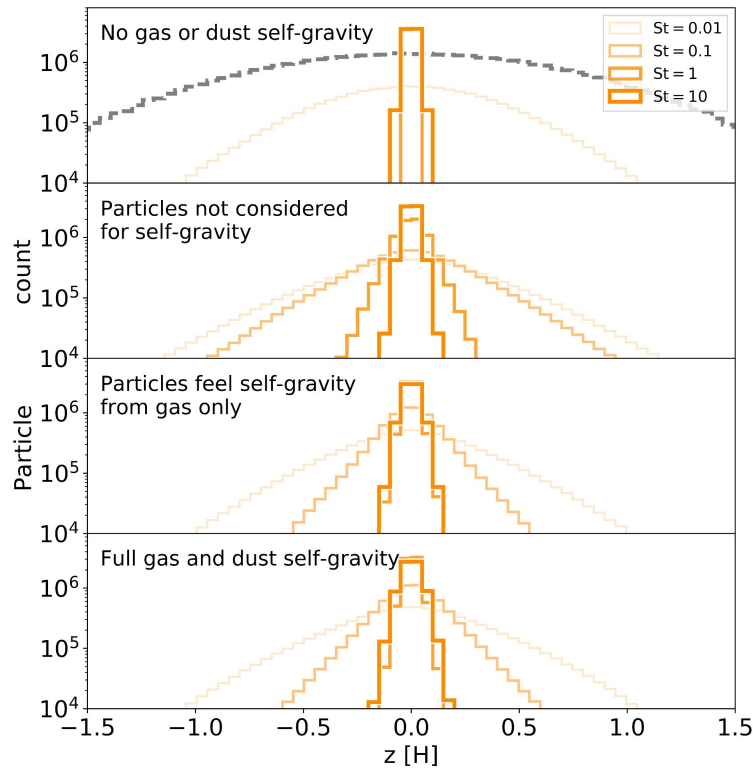
Efficient Settling

Settling versus turbulent diffusion

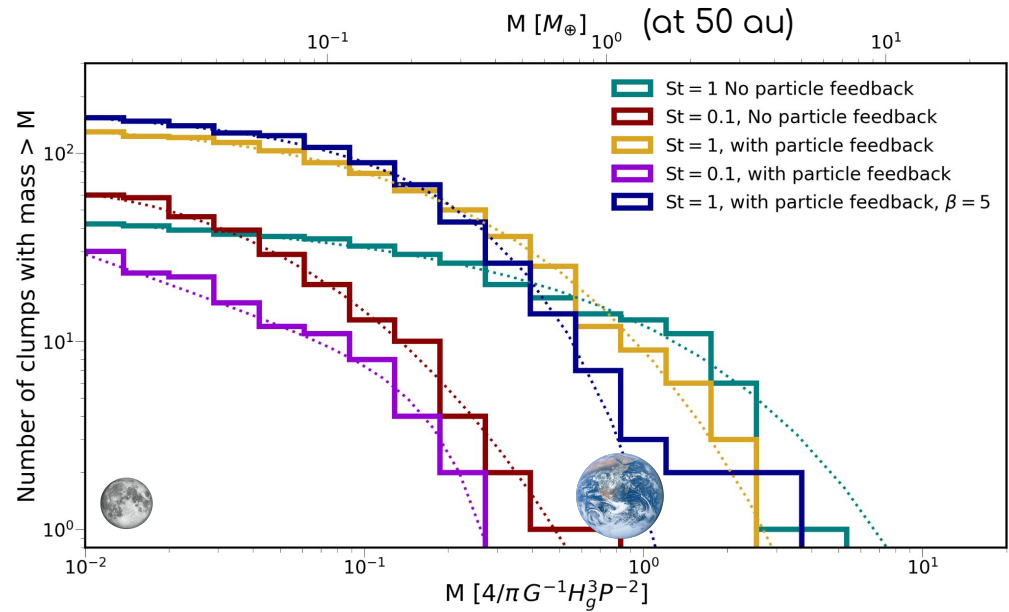
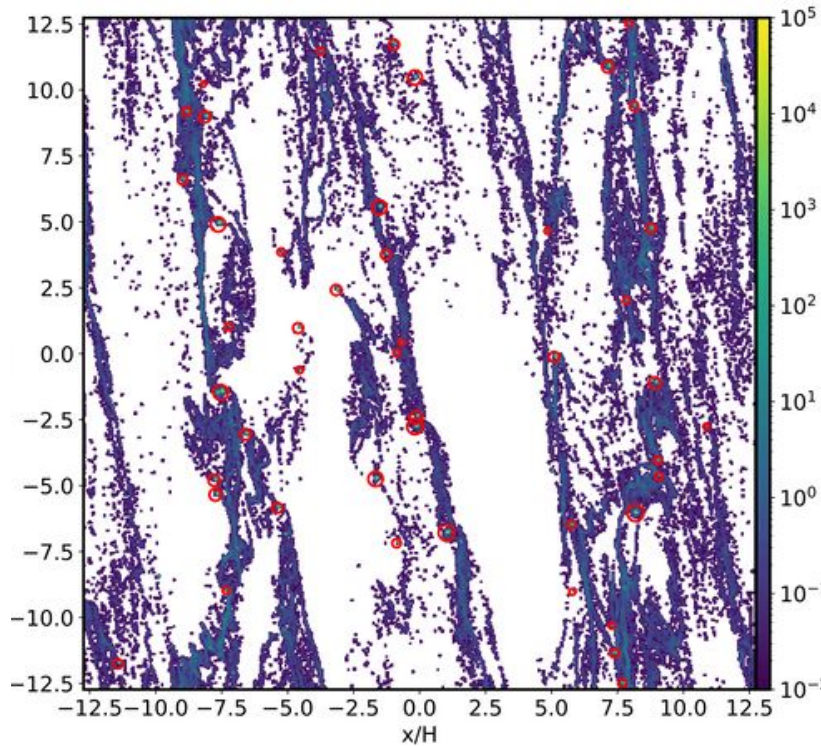
Dust layer is modified by self-gravity of the gas

Anisotropic dust diffusion:
thin dust layer even with
strong accretion

$Sc \sim 10$



Baehr & Zhu (2021b)

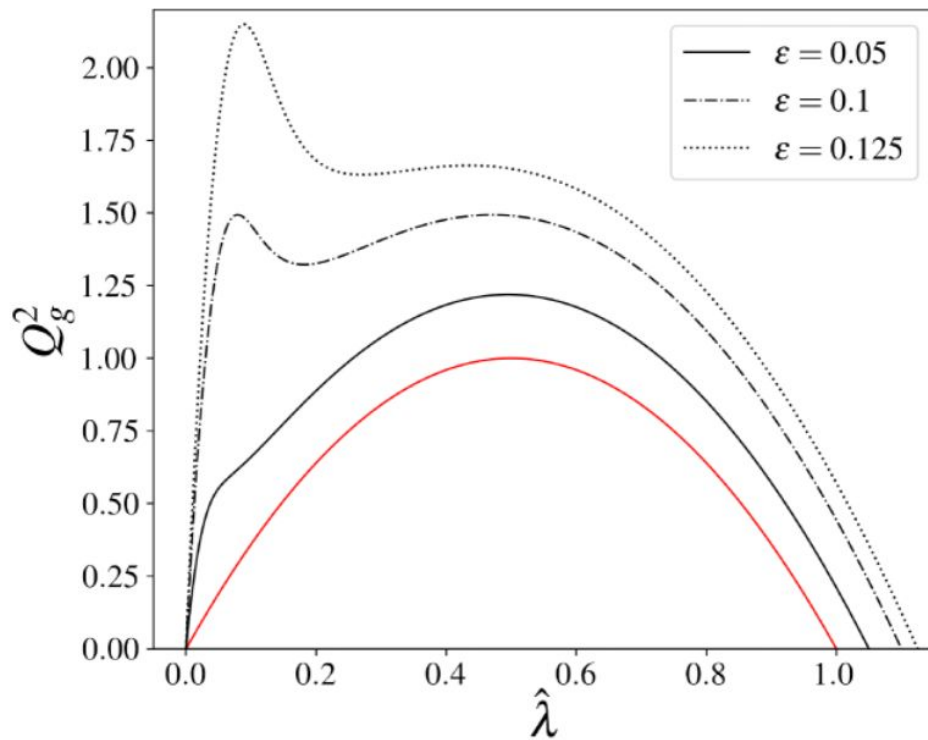


cores
Formation of planetesimals in a GI disk

Baehr, Zhu & Yang (2022)

Dust can gravitationally collapse even though the gas is stable

And the critical wavelength is significantly smaller



Longarini et al. (2023)

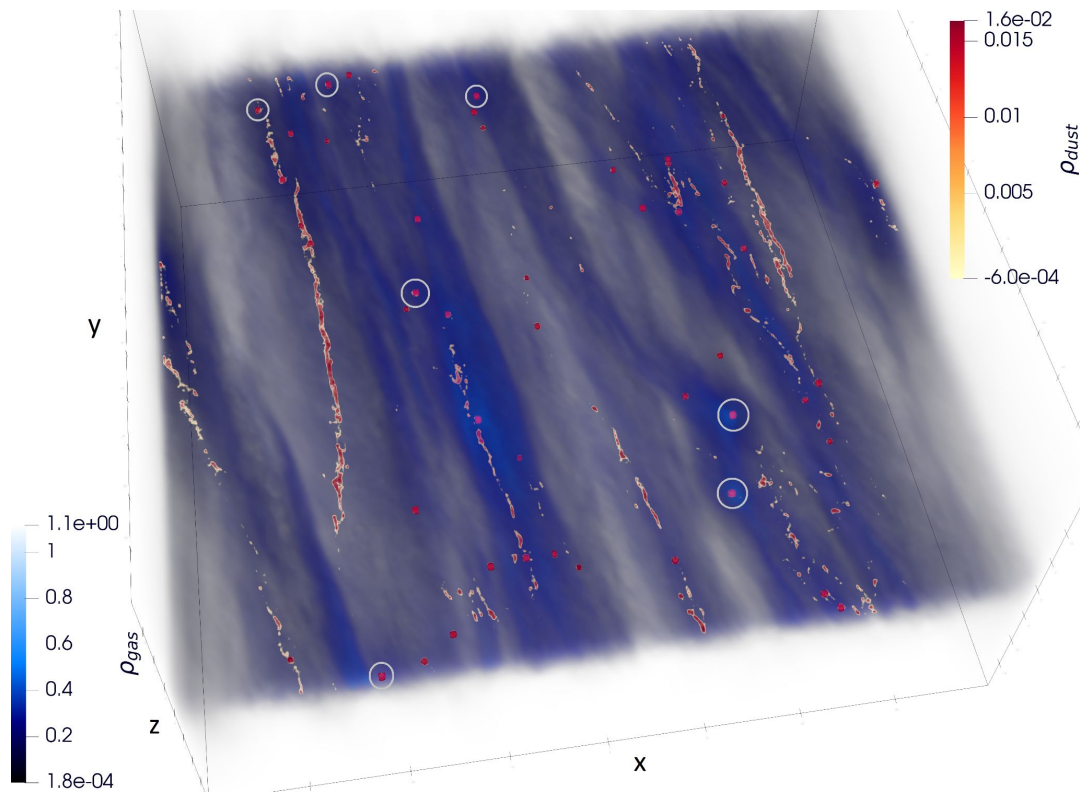
3d cloud:



l_c : critical size for a pebble cloud at Hill density to have gravity overcome turbulent diffusion

$$\frac{\delta}{St} \frac{c_s^2}{2\pi G \Sigma_{p,c}} < r < \frac{2\pi}{9} \frac{G \Sigma_{p,c}}{\Omega^2}$$

$$Q_d = \frac{3}{2} \frac{Q}{\epsilon Z} \sqrt{\frac{\delta}{St}}$$

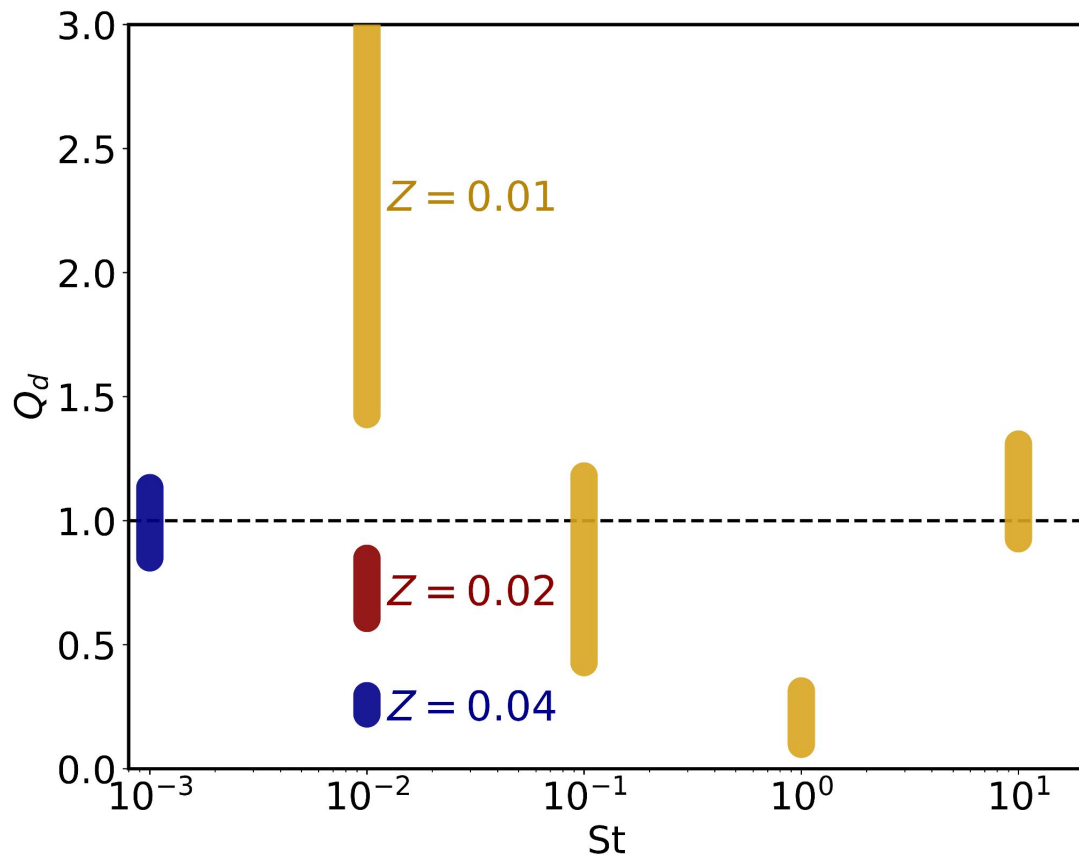


$$Q_d = \frac{3}{2} \frac{Q}{\epsilon Z} \sqrt{\frac{\delta}{\text{St}}}$$

Gerbig et al. (2020); Klahr & Schreiber (2021)

Dust concentrates and gravitationally collapses

Baehr et al. (2022)



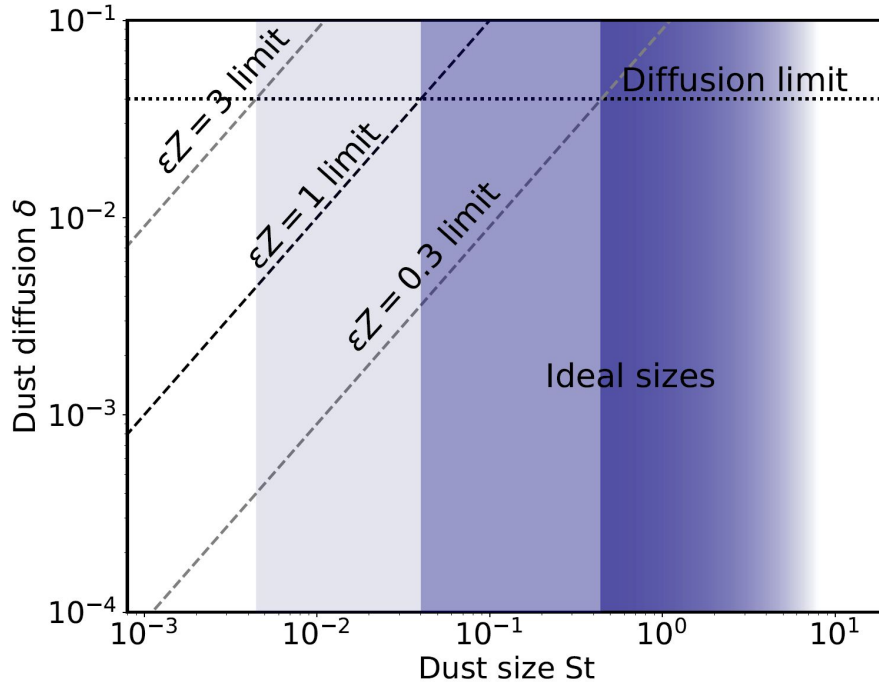
$$Q_d = \frac{3}{2} \frac{Q}{\epsilon Z} \sqrt{\frac{\delta}{St}}$$

Gerbig et al. (2020); Klahr & Schreiber (2021)

Dust concentrates and gravitationally collapses

Baehr et al. (2022)

What is the minimum dust size?

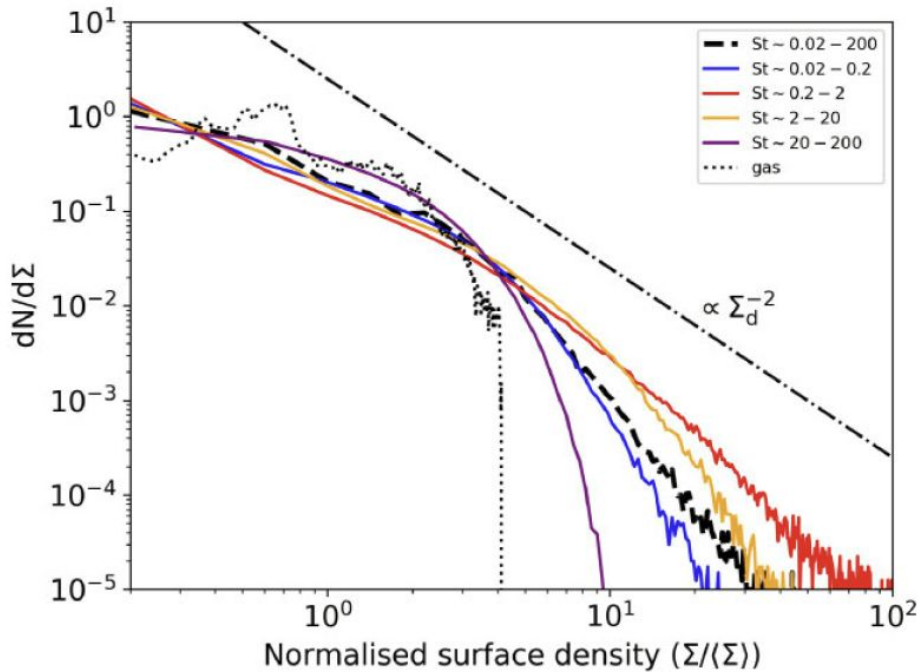
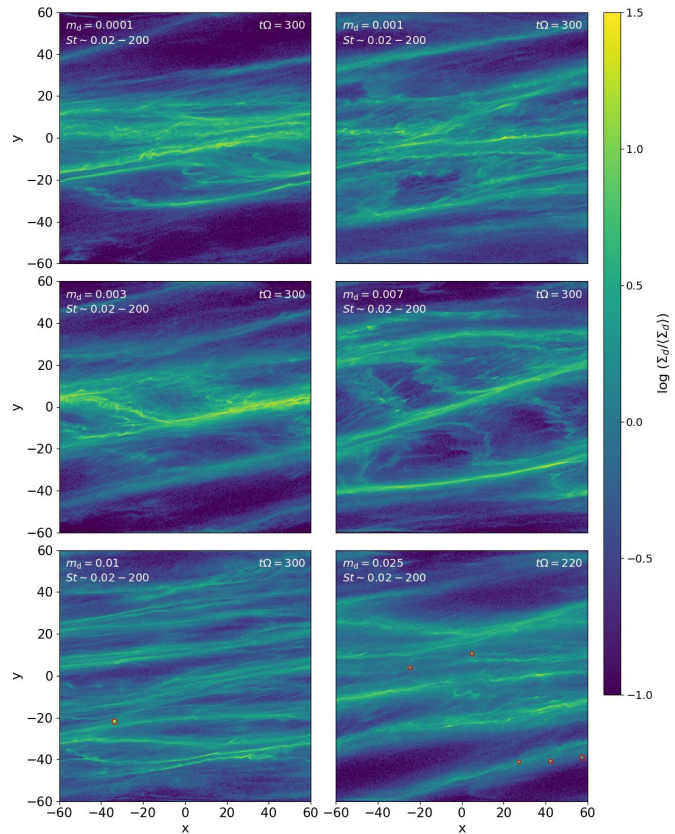


$$Q_d = \frac{3}{2} \frac{Q}{\epsilon Z} \sqrt{\frac{\delta}{St}}$$

Gerbig et al. (2020); Klahr & Schreiber (2021)

Self-gravity overcomes the diffusive motions due to turbulence

Baehr et al. (2025)

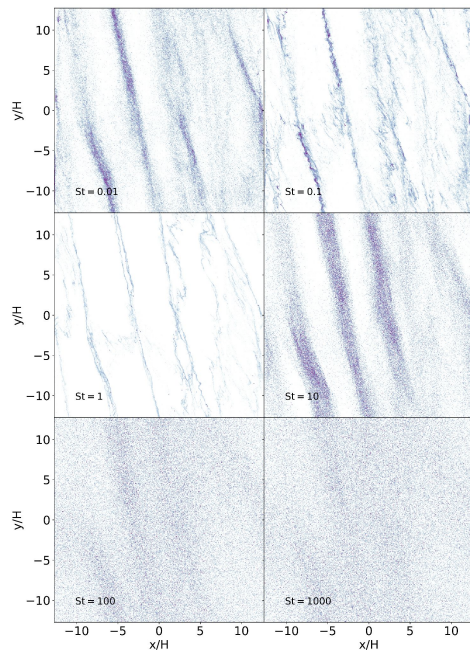


A realistic dust size distribution at ISM metallicity

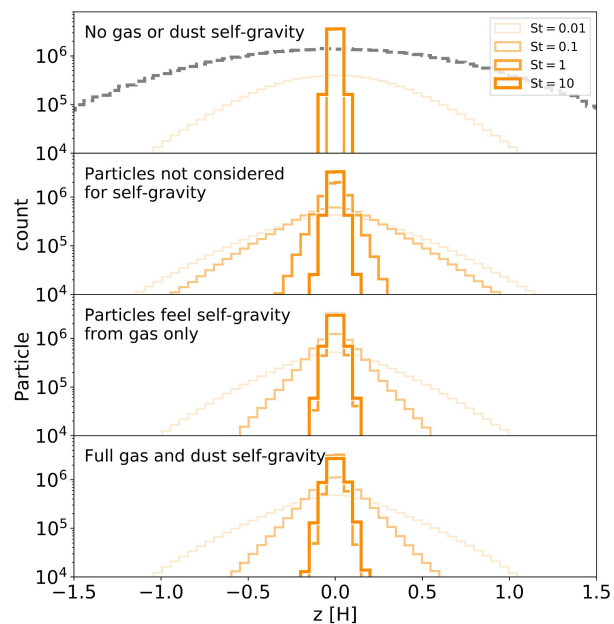
Rice, Baehr, et al. (2025)

In Short:

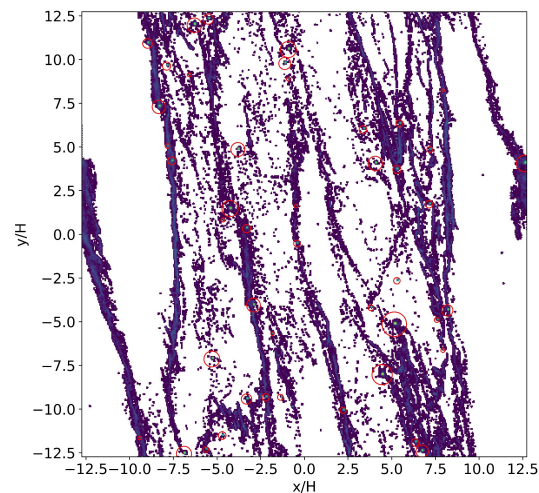
Efficient drift



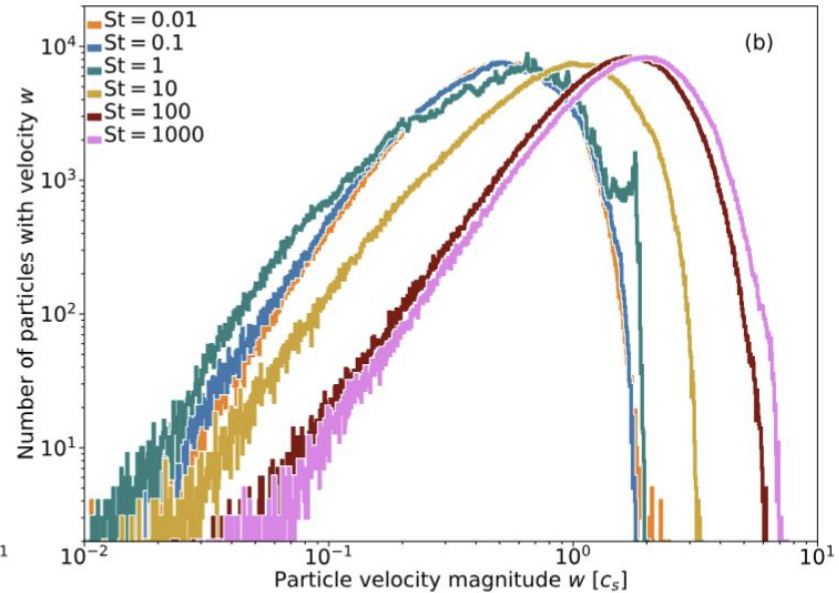
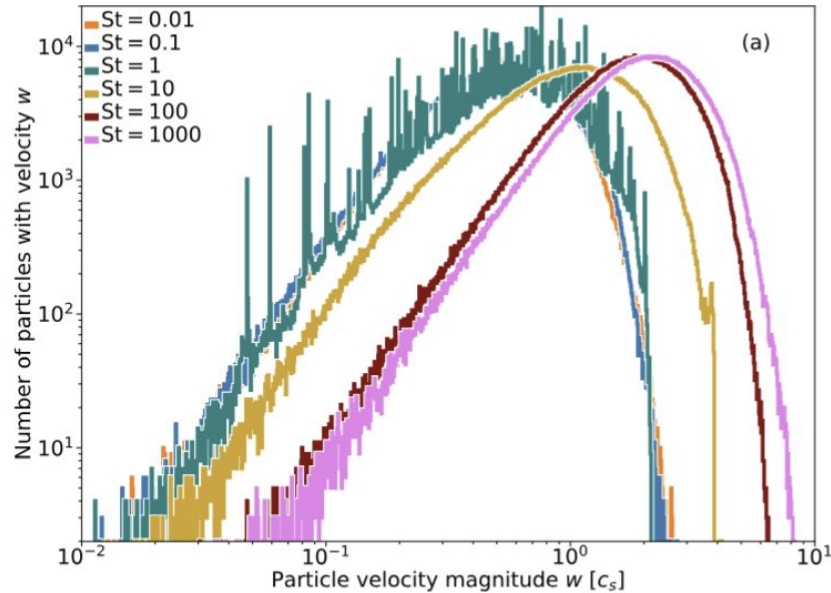
More settling



Dust concentration



(Backup slide) Effect of particle self-gravity



(Backup slide) Effect of resolution and initiation

